AI Grand Prix Virtual Qualifier Technical Specification

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1. Document Control

1.1 Revision History

Issue	Date	Author	Summary of Changes
00.01	2026-03-09	KH	First release

1.2 Audience

This specification is addressed to competition participants (“Teams”). It defines the technical interface and operational requirements required to develop autonomous control software for the Virtual AI Drone Race.

2. Purpose and Scope

2.1 Purpose

This document defines the interface between contestant-developed control software and the race simulator. The specification ensures that all teams interact with the simulator through a consistent communication and control interface.

2.2 Scope

- communication interfaces between contestant software and the simulator
- control input requirements
- telemetry interfaces
- vision data interfaces
- simulation timing and performance constraints
- virtual race environment definition
- qualification run requirements

2.3 Out of Scope

- internal simulator architecture
- event operations
- contractual or commercial matters

3. Simulation Environment

3.1 General Environment

The race takes place within a high-fidelity real-time physics simulator.

- start gate
- sequential race gates
- finish gate
- vertical and horizontal obstacles
- boundary elements
- terrain and environmental structures

Gates will be visually distinctive to the environment, but consistent throughout the Virtual Qualifier 1 track.

3.2 Physical Simulation Model

The simulator implements a rigid-body drone flight model including thrust generation, aerodynamic drag, gravity, and collision physics.
Target physics update frequency: 120 Hz.

3.3 Spatial Reference Model

The simulator uses a local Cartesian coordinate system internally. No geographic coordinates are provided. GPS simulation is not available and absolute global position is not exposed.

3.4 Visual Environment

The simulator provides a forward-facing first-person camera and visual environment including gates, course guidance structures, static scene objects, and dynamic lighting.

3.5 Environmental Determinism

- course geometry is identical for all participants
- physics parameters are identical
- environmental conditions are deterministic

4. Communication Protocol — MAVLink Interface

4.1 Overview

The simulator communicates with contestant control software using MAVLink v2 through MAVSDK-compatible interfaces.

4.2 Transport

Supported transport: UDP

4.3 Supported MAVLink Messages

Message	Direction	Purpose
HEARTBEAT	Simulator → Client	Connection status
ATTITUDE	Simulator → Client	Vehicle attitude
HIGHRES_IMU	Simulator → Client	Vehicle status
SET_POSITION_TARGET_LOCAL_NED	Client → Simulator	Control interface
SET_ATTITUDE_TARGET	Client → Simulator	Control interface
TIMESYNC	Simulator → Client	Timing
ODOMETRY		

4.4 Timing

Physics simulation rate: 120 Hz
Recommended command rate: 50–120 Hz
Minimum heartbeat rate: 2 Hz

4.5 Telemetry

- vehicle attitude
- orientation
- linear velocities
- system status flags
- simulator navigation reference data

4.6 Vision Stream

Vision stream parameters will be provided in a separate specification.

4.7 Software-in-the-Loop Bridge

The simulator provides a low-latency UDP SITL bridge enabling external AI controllers to exchange telemetry and control commands.

5. Contestant Software Environment

5.1 Runtime Environment

Participants may assume a Python-based runtime environment. Python 3.14.2 is known to operate correctly. Participants are allowed to choose other environments

5.2 Client Responsibilities

- establish MAVLink communication
- maintain heartbeat messages
- send control commands
- process telemetry data
- process vision stream data

5.3 Intended Control Architecture

Typical conceptual control pipeline:
Vision + Telemetry → Perception → Planning → Control → Pilot Commands → Stabilized Controller

6. Example Control Session

- Client initializes MAVSDK
- Client connects to simulator endpoint
- Simulator transmits HEARTBEAT
- Client streams control commands
- Simulator applies commands
- Telemetry and vision streams returned

7. Compliance

Participants must ensure their implementation conforms to this specification. Specifically, human interaction during the flight which the participants submit as a timed run is grounds for immediate disqualification.

8. Round One — Qualification Phase

8.1 Objective

Round One verifies that contestant software can successfully navigate the racecourse.

8.2 Course Structure

- start gate
- intermediate gates
- finish gate

8.3 Maximum Run Duration

Maximum run duration: 8 minutes.